

From Slices to Smoothed Slices

A Detailed Overleaf Report on the Formulation, Intuition, Implementation Path, and Research Program

Prepared from the working discussion on SLICES, NF-ISAM, and SMOOTHED SLICES

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1 Executive Summary

The purpose of this report is to document the full technical path that led from the original SLICES perspective and NF-iSAM toward the proposed SMOOTHED SLICES method. The most important idea is the conversion of a repeated nested sampling operation into the **smoothing property of conditional expectation**:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{G}]]. \quad (1)$$

The core interpretation is simple:

Core intuition

The original SLICES method uses graph structure to find sampleable factors. It then samples deeper variables and evaluates remaining factors along those samples. SMOOTHED SLICES uses the same graph-structural route, but it inserts the future factors into reusable conditional-expectation surfaces. These surfaces summarize the expected contribution of future factors, conditioned on the current variable and represented over the separator.

The most compact formulation of the proposed replacement is

$$\hat{f}_{\text{new,paper}}^{(2)}(x_2 \mid D_2) = c_1 \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) Z_l(x_1^{(n)}) \frac{1}{M_n} \sum_{m=1}^{M_n} f(l_1^{(n,m)}, x_2), \quad (2)$$

$$\hat{f}_{\text{new,smooth}}^{(2)}(x_2 \mid D_2) = c_1 \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) \hat{R}_1(x_1^{(n)}, x_2), \quad (3)$$

where the conditional smoothing surface is

$$R_1(x_1, x_2) = Z_l(x_1) \mathbb{E}[f(L_1, x_2) \mid X_1 = x_1] = \int f(x_1, l_1) f(l_1, x_2) dl_1. \quad (4)$$

The proposed method does not claim that the integrals disappear. Instead, it asks whether the conditional expectation surfaces such as R_1 and the recursive R_r can be represented with much smaller online cardinality than the raw nested particle tree. If yes, then the method may reduce variance, improve reuse, and reduce computational effort.

2 Reconstructed Discussion Path: How the Idea Emerged

This section reconstructs the conceptual path from the discussion. It is not a verbatim transcript of every message, but it preserves the technical sequence and the design decisions that converged to SMOOTHED SLICES.

2.1 Starting point: comparing NF-iSAM and Slices

The discussion began with two nonparametric SLAM inference methods:

- NF-iSAM: learns non-Gaussian Bayes-tree clique conditionals using normalizing flows, then samples the posterior by top-down ancestral sampling.
- SLICES: avoids KDEs and normalizing-flow training by directly accessing conditional slices of high-dimensional local factor products during variable elimination.

The important contrast was not merely implementation style. The contrast was about where the method pays computational cost:

NF-iSAM : learn low-dimensional clique conditionals with neural density models, (5)

SLICES : sample from available local structure and evaluate factor-product slices directly. (6)

2.2 First insight: Slices avoids learned intermediate reconstructions

The SLICES paper claims that one can approximate the new separator factors and conditionals directly from high-dimensional slices, without KDE or normalizing-flow reconstruction at each step. That suggested a useful implementation direction: build a demonstrator that keeps local factor products explicit and visualizes what is being sliced.

2.3 Second insight: the hidden proposal in the Slices estimator

We then noticed that the paper’s estimator samples the eliminated variable from a selected factor f' and removes that factor from the explicit product. This can look as if f' was ignored. It is not ignored; it is inside the sampling measure. This led to the explicit proposal notation

$$q_j(\theta_j) = \frac{f'(\theta_j)}{Z_{f'}}, \quad Z_{f'} = \int f'(\theta_j) d\theta_j. \quad (7)$$

This was important because it clarified that later improvements should preserve proposal correctness, not simply delete factors.

2.4 Third insight: the Eq. (16)-type nested step is the critical location

The key location became the paper’s example in which x_1 is eliminated first, creating an approximate factor over (l_1, x_2) . Then l_1 is eliminated although it has no unary factor. The paper uses the structure induced by the previous elimination to sample l_1 .

The expression corresponding to the paper’s Eq. (16) has the form

$$\tilde{f}_{\text{new}}^{(2)}(x_2 \mid D_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) \int f(x_1^{(n)}, l_1) f(l_1, x_2) dl_1. \quad (8)$$

The paper then estimates the integral by additional samples of l_1 . This was identified as the place where nested sampling enters.

2.5 Fourth insight: overlap-aware mixture slices

Before arriving at the full smoothing formulation, we discussed an intermediate idea: instead of sampling l_1 only from $f(x_1^{(n)}, l_1)$, use a proposal that sees the overlap with the future factor $f(l_1, x_2)$.

For each outer sample $x_1^{(n)}$, define

$$A_n(x_2) \triangleq f(x_1^{(n)}, x_2), \quad B_n(l_1) \triangleq f(x_1^{(n)}, l_1), \quad C(l_1, x_2) \triangleq f(l_1, x_2). \quad (9)$$

Then

$$\tilde{f}_{\text{new}}^{(2)}(x_2 \mid D_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} A_n(x_2) K_n(x_2), \quad (10)$$

$$K_n(x_2) = \int B_n(l_1) C(l_1, x_2) dl_1. \quad (11)$$

For a fixed x_2 , the zero-variance proposal would be

$$q_n^*(l_1 \mid x_2) = \frac{B_n(l_1) C(l_1, x_2)}{K_n(x_2)}. \quad (12)$$

For a proposal reusable across x_2 , with separator relevance density $\rho(x_2)$, a more practical target is

$$q_n^*(l_1) \propto B_n(l_1) \sqrt{\int \rho(x_2) C^2(l_1, x_2) dx_2}. \quad (13)$$

This idea already pointed toward the importance of future factors, but it was still mainly a better proposal design.

2.6 Fifth insight: retaining a mixture surface

The next step was to retain the full mixture surface instead of collapsing each sample independently:

$$H_N(l_1, x_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} A_n(x_2) B_n(l_1) C(l_1, x_2). \quad (14)$$

Then

$$\tilde{f}_{\text{new}}^{(2)}(x_2 \mid D_2) = \int H_N(l_1, x_2) dl_1. \quad (15)$$

A global proposal for integrating over l_1 could be

$$q^*(l_1) \propto \sqrt{\int \rho(x_2) H_N^2(l_1, x_2) dx_2}. \quad (16)$$

This was useful but still left us with the question: can we do more than choose a better proposal? Can we summarize future contributions themselves?

2.7 Sixth insight: the smoothing property of conditional expectation

The central conceptual jump was to read Eq. (8) as a conditional expectation rather than as merely a nested Monte Carlo integral. For fixed x_2 , define

$$q_l(l_1 \mid x_1) = \frac{f(x_1, l_1)}{Z_l(x_1)}, \quad Z_l(x_1) = \int f(x_1, l_1) dl_1. \quad (17)$$

Then

$$\int f(x_1, l_1) f(l_1, x_2) dl_1 = Z_l(x_1) \mathbb{E}[f(L_1, x_2) \mid X_1 = x_1]. \quad (18)$$

This directly suggested the surface $R_1(x_1, x_2)$ in Eq. (4). The method became SMOOTHED SLICES: use the same structural sampling path as SLICES, but replace repeated nested sampling by conditional smoothing surfaces.

2.8 Seventh insight: recursive surfaces for deeper chains

After defining R_1 , we extended the same idea to longer paths. For a path

$$\xi_0 \rightarrow \xi_1 \rightarrow \xi_2 \rightarrow \cdots \rightarrow \xi_H, \quad (19)$$

we define recursive smoothing surfaces $R_r(\xi_r, s)$ that summarize all future factors downstream of ξ_r toward the separator s :

$$R_r(\xi_r, s) = \int g_r(\xi_r, \xi_{r+1}) a_r(\xi_{r+1}, s) R_{r+1}(\xi_{r+1}, s) d\xi_{r+1}. \quad (20)$$

This became the general RECURSIVE CONDITIONAL-SMOOTHING SLICES formulation.

2.9 Eighth insight: implementation must be finite-cardinality

The method only becomes computationally useful if R_r can be represented compactly. Therefore we introduced support spaces

$$\mathcal{X}_r = \{\xi_r^{(1)}, \dots, \xi_r^{(B_r)}\}, \quad |\mathcal{X}_r| = B_r, \quad (21)$$

$$\mathcal{S} = \{s^{(1)}, \dots, s^{(R_s)}\}, \quad |\mathcal{S}| = R_s. \quad (22)$$

The surface is stored as

$$\mathbf{R}_r \in \mathbb{R}^{B_r \times R_s}, \quad [\mathbf{R}_r]_{a,\rho} = \hat{R}_r(\xi_r^{(a)}, s^{(\rho)}). \quad (23)$$

2.10 Ninth insight: the open research question

The final open question became: **how many computations can this actually reduce?** The careful answer is that this is not guaranteed a priori. The method reduces complexity only if the surfaces are smoother, sparse, low-rank, compressible, or representable with small active successor sets.

The comparison is

$$\text{paper-like nested work} \sim N^H R_s, \quad (24)$$

$$\text{smooth-surface work} \sim \sum_{r=0}^{H-1} B_r K_r R_s, \quad (25)$$

where K_r is the number of active successors per support point. The method is computationally attractive when

$$\sum_{r=0}^{H-1} B_r K_r R_s \ll N^H R_s. \quad (26)$$

3 The Original Slices Method

3.1 Factor graph and posterior factorization

Let

$$G = (F, \Theta, E) \quad (27)$$

be a factor graph, where F is the set of factor nodes, Θ is the set of random variables, and E is the set of edges. The posterior is represented by

$$P(\Theta \mid D) \propto \prod_i f_i(\Theta_i), \quad (28)$$

where $\Theta_i \subseteq \Theta$ denotes the variables adjacent to factor f_i .

The SLICES paper assumes unary and pairwise factors for simplicity, but the algebra is easiest to write using the general product notation.

3.2 Exact forward elimination

Let the elimination ordering be

$$O = \{\theta_1, \theta_2, \dots, \theta_{|\Theta|}\}. \quad (29)$$

Before eliminating θ_j , let the reduced graph be

$$G_{j-1} = (F_{j-1}, \Theta_{j-1}, E_{j-1}). \quad (30)$$

The removed factor set is

$$F_{j-1}(\theta_j) = \{f \in F_{j-1} : \theta_j \in \Theta_f\}. \quad (31)$$

The separator is the set of all variables attached to those factors except θ_j :

$$S_j = \left(\bigcup_{f_i \in F_{j-1}(\theta_j)} \Theta_i \right) \setminus \{\theta_j\}. \quad (32)$$

The local joint density is

$$P_{\text{joint}}(\theta_j, S_j \mid D_j) = \eta_j^{-1} \prod_{f_i \in F_{j-1}(\theta_j)} f_i(\Theta_i). \quad (33)$$

By the chain rule,

$$P_{\text{joint}}(\theta_j, S_j \mid D_j) = P(\theta_j \mid S_j, D_j) f_{\text{new}}(S_j \mid D_j). \quad (34)$$

The new separator factor is the marginal

$$f_{\text{new}}(S_j \mid D_j) = \int P_{\text{joint}}(\theta_j, S_j \mid D_j) d\theta_j \quad (35)$$

$$= \eta_j^{-1} \int \prod_{f_i \in F_{j-1}(\theta_j)} f_i(\Theta_i) d\theta_j. \quad (36)$$

3.3 Slice Monte Carlo estimator

The SLICES estimator chooses a factor $f' \in F_{j-1}(\theta_j)$ from which samples can be generated. If

$$q_j(\theta_j) = \frac{f'(\theta_j)}{Z_{f'}}, \quad Z_{f'} = \int f'(\theta_j) d\theta_j, \quad (37)$$

then Eq. (36) can be written as

$$f_{\text{new}}(S_j \mid D_j) = \eta_j^{-1} Z_{f'} \mathbb{E}_{\theta_j \sim q_j} \left[\prod_{f_i \in F_{j-1}(\theta_j) \setminus \{f'\}} f_i(\Theta_i) \right]. \quad (38)$$

With samples $\theta_j^{(n)} \sim q_j$, the estimator is

$$\hat{f}_{\text{new}}(S_j \mid D_j) = \frac{\eta_j^{-1} Z_{f'}}{N_j} \sum_{n=1}^{N_j} \prod_{f_i \in F_{j-1}(\theta_j) \setminus \{f'\}} f_i(\theta_j^{(n)}, \Theta_i \setminus \{\theta_j\}). \quad (39)$$

Why this is called slices

Each sample $\theta_j^{(n)}$ fixes one coordinate of the high-dimensional local joint surface. The remaining variables, $\Theta_i \setminus \{\theta_j\}$, form a conditional slice. The estimator averages such slices to approximate the marginal over the separator.

3.4 The conditional estimator

From Eq. (34), the conditional is

$$P(\theta_j \mid S_j, D_j) = \frac{P_{\text{joint}}(\theta_j, S_j \mid D_j)}{f_{\text{new}}(S_j \mid D_j)}. \quad (40)$$

Replacing f_{new} by the slice estimator gives

$$\hat{P}(\theta_j \mid S_j, D_j) = \frac{\prod_{f_i \in F_{j-1}(\theta_j)} f_i(\Theta_i)}{\frac{1}{N_j} \sum_{n=1}^{N_j} \prod_{f_i \in F_{j-1}(\theta_j) \setminus \{f'\}} f_i(\theta_j^{(n)}, \Theta_i \setminus \{\theta_j\})}, \quad (41)$$

where global normalizing constants are typically absorbed or cancel.

3.5 Backward pass

Once the Bayes net conditionals are constructed, marginals are recovered in reverse elimination order:

$$P(\theta_j \mid D) = \int P(\theta_j \mid S_j, D_j) P(S_j \mid D) dS_j. \quad (42)$$

With samples $S_j^{(n)} \sim \hat{P}(S_j \mid D)$,

$$\hat{P}(\theta_j \mid D) = \frac{1}{N} \sum_{n=1}^N \hat{P}(\theta_j \mid S_j^{(n)}, D_j). \quad (43)$$

The joint posterior can also be sampled by traversing the Bayes net in reverse elimination order and sampling each variable from its conditional given a separator realization.

3.6 Lemma-1 sampleability

The SLICES paper’s Lemma 1 is crucial. It says, in effect, that if each eliminated variable either has a unary factor or is connected to a previously eliminated variable under a suitable ordering,

then samples of the current variable can be drawn from one of the factors in the current reduced graph.

Implementation consequence

A Slices implementation must maintain, at each elimination step, a record of which factor is used as the sampling factor. If no unary factor is available, it must search for a factor induced by a previous elimination that contains the current variable in a sampleable unary-like form after conditioning on previously sampled values.

4 The Critical Example: From x_1 to l_1 to x_2

4.1 First elimination

Consider the local graph containing

$$f(x_1), \quad f(x_1, x_2), \quad f(x_1, l_1), \quad f(l_1, x_2). \quad (44)$$

The elimination order begins with

$$\theta_1 = x_1, \quad \theta_2 = l_1. \quad (45)$$

Eliminating x_1 first gives separator

$$S_1 = \{l_1, x_2\}. \quad (46)$$

The exact new factor is

$$f_{\text{new}}^{(1)}(l_1, x_2 \mid D_1) = \eta_1^{-1} \int f(x_1) f(x_1, x_2) f(x_1, l_1) dx_1. \quad (47)$$

Sampling from

$$q_1(x_1) = \frac{f(x_1)}{Z_1}, \quad Z_1 = \int f(x_1) dx_1, \quad (48)$$

we get

$$\hat{f}_{\text{new}}^{(1)}(l_1, x_2 \mid D_1) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) f(x_1^{(n)}, l_1), \quad (49)$$

where $c_1 = \eta_1^{-1} Z_1$.

4.2 Second elimination

Now eliminate l_1 . The important detail is that D_2 contains the approximate factor generated at the previous step:

$$D_2 = \{\hat{f}_{\text{new}}^{(1)}(l_1, x_2 \mid D_1), z_2\}. \quad (50)$$

The local joint is

$$P_{\text{joint}}(l_1, x_2 \mid D_2) = \hat{f}_{\text{new}}^{(1)}(l_1, x_2 \mid D_1) f(l_1, x_2). \quad (51)$$

Substituting Eq. (49) into Eq. (51) and marginalizing l_1 gives

$$\tilde{f}_{\text{new}}^{(2)}(x_2 \mid D_2) = \int \tilde{f}_{\text{new}}^{(1)}(l_1, x_2 \mid D_1) f(l_1, x_2) dl_1 \quad (52)$$

$$= \int \left[\frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) f(x_1^{(n)}, l_1) \right] f(l_1, x_2) dl_1 \quad (53)$$

$$= \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) \int f(x_1^{(n)}, l_1) f(l_1, x_2) dl_1. \quad (54)$$

Eq. (54) is the critical expression. The paper's nested sampling step estimates the inner integral separately for each $x_1^{(n)}$.

4.3 The paper-style nested estimator

For each $x_1^{(n)}$, define

$$q_l(l_1 \mid x_1^{(n)}) = \frac{f(x_1^{(n)}, l_1)}{Z_l(x_1^{(n)})}, \quad Z_l(x_1^{(n)}) = \int f(x_1^{(n)}, l_1) dl_1. \quad (55)$$

Then

$$\int f(x_1^{(n)}, l_1) f(l_1, x_2) dl_1 = Z_l(x_1^{(n)}) \mathbb{E}[f(L_1, x_2) \mid X_1 = x_1^{(n)}] \quad (56)$$

$$\approx Z_l(x_1^{(n)}) \frac{1}{M_n} \sum_{m=1}^{M_n} f(l_1^{(n,m)}, x_2). \quad (57)$$

Therefore,

$$\hat{f}_{\text{new,paper}}^{(2)}(x_2 \mid D_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) Z_l(x_1^{(n)}) \frac{1}{M_n} \sum_{m=1}^{M_n} f(l_1^{(n,m)}, x_2). \quad (58)$$

Where the nested cost enters

The term $l_1^{(n,m)}$ is conditional on $x_1^{(n)}$. If the same pattern continues through more variables, the implementation can create nested sample families of the form $(n, m, k, \dots)$.

5 The Smoothing-Property Conversion

5.1 Step 1: define the random variable being estimated

For fixed separator value x_2 , define

$$Y_{x_2} = f(X_1, x_2) Z_l(X_1) f(L_1, x_2), \quad (59)$$

where

$$X_1 \sim q_1(x_1), \quad L_1 \mid X_1 = x_1 \sim q_l(l_1 \mid x_1). \quad (60)$$

Then the exact target in Eq. (54) is proportional to

$$\mathbb{E}[Y_{x_2}] = \mathbb{E}_{X_1}[f(X_1, x_2) Z_l(X_1) \mathbb{E}[f(L_1, x_2) \mid X_1]]. \quad (61)$$

5.2 Step 2: define the conditional smoothing surface

Define

$$R_1(x_1, x_2) \triangleq Z_l(x_1) \mathbb{E}[f(L_1, x_2) \mid X_1 = x_1] \quad (62)$$

$$= \int f(x_1, l_1) f(l_1, x_2) dl_1. \quad (63)$$

Then

$$\mathbb{E}[Y_{x_2}] = \mathbb{E}_{X_1}[f(X_1, x_2) R_1(X_1, x_2)]. \quad (64)$$

Consequently, the Smoothed Slices estimator is

$$\hat{f}_{\text{new,smooth}}^{(2)}(x_2 \mid D_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) \hat{R}_1(x_1^{(n)}, x_2). \quad (65)$$

What R_1 means

$R_1(x_1, x_2)$ is not a posterior density and not a point estimate. It is a conditional expectation surface. It answers the question: if the current retained path value is x_1 , what is the expected contribution of the future landmark variable l_1 to every possible separator value x_2 ?

5.3 Step 3: state the smoothing property explicitly

Eq. (61) is exactly the smoothing property:

$$\mathbb{E}[Y_{x_2}] = \mathbb{E}[\mathbb{E}[Y_{x_2} \mid X_1]]. \quad (66)$$

The conditional expectation is

$$\mathbb{E}[Y_{x_2} \mid X_1 = x_1] = f(x_1, x_2) R_1(x_1, x_2). \quad (67)$$

Thus, the Smoothed Slices estimator keeps the outer samples $x_1^{(n)}$ but replaces the inner empirical conditional average over l_1 by a reusable approximation $\hat{R}_1$.

5.4 Step 4: variance interpretation

The law of total variance gives

$$\text{Var}(Y_{x_2}) = \text{Var}(\mathbb{E}[Y_{x_2} \mid X_1]) + \mathbb{E}[\text{Var}(Y_{x_2} \mid X_1)]. \quad (68)$$

Therefore,

$$\text{Var}(\mathbb{E}[Y_{x_2} \mid X_1]) \leq \text{Var}(Y_{x_2}). \quad (69)$$

Variance intuition

Sampling L_1 explicitly injects conditional noise. Replacing the random future contribution by its conditional expectation removes that conditional noise if R_1 is exact. If R_1 is approximate, the method trades inner Monte Carlo noise for surface approximation error.

For the paper-style nested estimator, a schematic variance decomposition is

$$\text{Var}[\hat{f}_{\text{new,paper}}^{(2)}(x_2 \mid D_2)] = \frac{c_1^2}{N_1} \text{Var}[f(X_1, x_2) R_1(X_1, x_2)] \quad (70)$$

$$+ \frac{c_1^2}{N_1} \mathbb{E} \left[\frac{f^2(X_1, x_2) Z_l^2(X_1)}{M(X_1)} \text{Var}(f(L_1, x_2) \mid X_1) \right]. \quad (71)$$

The second term is the extra nested sampling noise targeted by SMOOTHED SLICES.

6 Recursive Conditional-Smoothing Slices

6.1 Path abstraction

For a longer structurally sampleable path, write

$$\xi_0 \rightarrow \xi_1 \rightarrow \cdots \rightarrow \xi_H. \quad (72)$$

Here:

- ξ_r is the r -th variable along the structural sampling path.
- H is the path depth.
- s is the separator value over which the final new factor is represented.
- $g_r(\xi_r, \xi_{r+1})$ is the sampling factor that makes ξ_{r+1} sampleable once ξ_r is fixed.
- $a_r(\xi_{r+1}, s)$ is the future factor contribution that should be fused toward the separator.
- $R_r(\xi_r, s)$ is the recursive smoothing surface summarizing downstream factors after level r .

Normalize each sampling factor:

$$q_r(\xi_{r+1} \mid \xi_r) = \frac{g_r(\xi_r, \xi_{r+1})}{Z_r(\xi_r)}, \quad Z_r(\xi_r) = \int g_r(\xi_r, \lambda) d\lambda. \quad (73)$$

Initialize the deepest surface by

$$R_H(\xi_H, s) = a_H(\xi_H, s). \quad (74)$$

For $r = H - 1, H - 2, \dots, 0$, define

$$R_r(\xi_r, s) \triangleq Z_r(\xi_r) \mathbb{E}_{\xi_{r+1} \sim q_r(\cdot \mid \xi_r)} [a_r(\xi_{r+1}, s) R_{r+1}(\xi_{r+1}, s)] \quad (75)$$

$$= \int g_r(\xi_r, \xi_{r+1}) a_r(\xi_{r+1}, s) R_{r+1}(\xi_{r+1}, s) d\xi_{r+1}. \quad (76)$$

The new separator factor is then approximated by

$$\hat{f}_{\text{new,smooth}}(s) = \frac{c_0}{N_0} \sum_{n=1}^{N_0} b_0(\xi_0^{(n)}, s) \hat{R}_0(\xi_0^{(n)}, s), \quad (77)$$

where b_0 contains the front-level factors connecting ξ_0 directly to the separator.

6.2 Tower-property derivation

Define information sets

$$\mathcal{G}_r = \sigma(\xi_0, \xi_1, \dots, \xi_r), \quad r = 0, 1, \dots, H. \quad (78)$$

Let the total future contribution for fixed s be

$$Y_s = b_0(\xi_0, s) \prod_{r=0}^{H-1} [Z_r(\xi_r) a_r(\xi_{r+1}, s)] a_H(\xi_H, s). \quad (79)$$

Then repeated use of Eq. (1) gives

$$\mathbb{E}[Y_s] = \mathbb{E}[\mathbb{E}[Y_s \mid \mathcal{G}_0]] \quad (80)$$

$$= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y_s \mid \mathcal{G}_1] \mid \mathcal{G}_0]] \quad (81)$$

$$= \mathbb{E}[\mathbb{E}[\cdots \mathbb{E}[Y_s \mid \mathcal{G}_{H-1}] \cdots \mid \mathcal{G}_0]]. \quad (82)$$

The surfaces R_r are finite representations of these nested conditional expectations.

6.3 Finite-cardinality implementation

Define support spaces

$$\mathcal{X}_r = \{\xi_r^{(1)}, \dots, \xi_r^{(B_r)}\}, \quad |\mathcal{X}_r| = B_r, \quad (83)$$

$$\mathcal{S} = \{s^{(1)}, \dots, s^{(R_s)}\}, \quad |\mathcal{S}| = R_s. \quad (84)$$

Store the surface as

$$\mathbf{R}_r \in \mathbb{R}^{B_r \times R_s}, \quad [\mathbf{R}_r]_{a,\rho} = \hat{R}_r(\xi_r^{(a)}, s^{(\rho)}). \quad (85)$$

The storage cardinality is

$$\text{storage}(\mathbf{R}_r) = |\mathcal{X}_r| |\mathcal{S}| = B_r R_s. \quad (86)$$

Define a conditional smoothing matrix and fusion tensor:

$$\mathbf{P}_r \in \mathbb{R}^{B_r \times B_{r+1}}, \quad [\mathbf{P}_r]_{a,b} \approx \mathbb{P}(\xi_{r+1} = \xi_{r+1}^{(b)} \mid \xi_r = \xi_r^{(a)}), \quad (87)$$

$$[\mathbf{A}_r]_{a,b,\rho} \triangleq Z_r(\xi_r^{(a)}) a_r(\xi_{r+1}^{(b)}, s^{(\rho)}). \quad (88)$$

The dense discrete recursion is

$$[\mathbf{R}_r]_{a,\rho} = \sum_{b=1}^{B_{r+1}} [\mathbf{P}_r]_{a,b} [\mathbf{A}_r]_{a,b,\rho} [\mathbf{R}_{r+1}]_{b,\rho}. \quad (89)$$

The dense cost is

$$\text{cost}_r^{\text{dense}} \sim |\mathcal{X}_r| |\mathcal{X}_{r+1}| |\mathcal{S}| = B_r B_{r+1} R_s. \quad (90)$$

For online inference, use an active successor set

$$\mathcal{N}_r(a) \subseteq \{1, \dots, B_{r+1}\}, \quad |\mathcal{N}_r(a)| = K_r, \quad K_r \ll B_{r+1}. \quad (91)$$

The sparse recursion is

$$[\mathbf{R}_r]_{a,\rho} \approx \sum_{b \in \mathcal{N}_r(a)} [\mathbf{P}_r]_{a,b} [\mathbf{A}_r]_{a,b,\rho} [\mathbf{R}_{r+1}]_{b,\rho}. \quad (92)$$

The sparse cost is

$$\text{cost}_r^{\text{sparse}} \sim |\mathcal{X}_r| |\mathcal{N}_r| |\mathcal{S}| = B_r K_r R_s. \quad (93)$$

Implementation meaning

The MATLAB backend should not store nested cells like `L{n}{m}{k}` whenever possible. It should store surface matrices `R{r}`, sparse successor index lists `Nidx{r,a}`, transition weights `P{r}`, and factor-evaluation tensors or on-demand tensor evaluators `A{r}`.

7 How Smoothed Slices Differs from Slices in Implementation

7.1 Conceptual difference

The SLICES implementation follows the pattern

sampleable factor $\rightarrow$ draw samples $\rightarrow$ evaluate remaining factors $\rightarrow$ repeat at next elimination. (94)

The SMOOTHED SLICES implementation follows the pattern

sampleable factor $\rightarrow$ insert future factors into $R_r \rightarrow$ reuse cached surfaces $\rightarrow$ sample only where still needed. (95)

7.2 Data structures

Object	SLICES implementation	SMOOTHED SLICES implementation
Elimination record	Stores removed factors, separator, sampling factor, samples	Same, plus path variables and surface dependencies
Samples	Flat samples per elimination and nested samples when required	Outer samples plus finite supports $\mathcal{X}_r$ and $\mathcal{S}$
Future factors	Evaluated directly inside nested sums	Absorbed into $R_r(\xi_r, s)$ surfaces
Intermediate new factor	Mixture of slices over separator	Mixture of smoothed slices over separator
Caching	Cached marginals and conditionals for backward pass and early stop	Cached marginals, conditionals, and smoothing surfaces
Complexity diagnostics	Number of samples, factor evaluations, runtime	Same plus B_r , K_r , R_s , surface error, reuse ratio, compression ratio

Table 1: Implementation differences between SLICES and SMOOTHED SLICES.

7.3 Algorithmic skeleton

Algorithm: one affected Smoothed Slices chain

1. Identify the affected elimination segment after a new measurement.
2. Build the structural path $\xi_0 \rightarrow \dots \rightarrow \xi_H$ using the same Lemma-1 sampleability logic as SLICES.
3. Build finite support spaces $\mathcal{X}_r$ and separator support $\mathcal{S}$.
4. Initialize $\mathbf{R}_H$ from the deepest factor contribution a_H .
5. For $r = H - 1, H - 2, \dots, 0$:
 - (a) construct active successor sets $\mathcal{N}_r(a)$;
 - (b) estimate transition weights $[\mathbf{P}_r]_{a,b}$;
 - (c) evaluate factor contribution tensor entries $[\mathbf{A}_r]_{a,b,\rho}$;
 - (d) update $\mathbf{R}_r$ using Eq. (92);
 - (e) optionally compress $\mathbf{R}_r$ using basis, low-rank, kernel, or regression models.
6. Estimate the new separator factor using Eq. (77).
7. Store $\hat{f}_{\text{new}}$, $\hat{P}(\theta_j | S_j, D_j)$, and $\mathbf{R}_r$ in the cache.
8. Export diagnostics: factor evaluations, runtime, surface storage, active successor counts, MMD/RMSE, and surface error proxies.

8 Compression, Basis, and Regression Options for R_r

8.1 Option A: direct finite-support update

The closest implementation to the original SLICES spirit is to use only finite samples and factor evaluations:

$$[\mathbf{R}_r]_{a,\rho} \leftarrow \sum_{b \in \mathcal{N}_r(a)} [\mathbf{P}_r]_{a,b} [\mathbf{A}_r]_{a,b,\rho} [\mathbf{R}_{r+1}]_{b,\rho}. \quad (96)$$

This option is easiest to debug and should be implemented first.

8.2 Option B: basis over the separator

Let $\{\phi_k(s)\}_{k=1}^{K_s}$ be a basis over separator values. Approximate

$$R_r(\xi_r, s) \approx \sum_{k=1}^{K_s} \alpha_{r,k}(\xi_r) \phi_k(s), \quad |\Phi_s| = K_s, \quad K_s \ll R_s. \quad (97)$$

Storage is reduced from

$$B_r R_s \quad \text{to} \quad B_r K_s. \quad (98)$$

This is important if the surface is smooth in separator space.

8.3 Option C: conditional mean regression

Generate training pairs

$$\mathcal{D}_r^{\text{train}} = \{(\xi_r^{(i)}, \xi_{r+1}^{(i)})\}_{i=1}^{B_r^{\text{train}}}, \quad |\mathcal{D}_r^{\text{train}}| = B_r^{\text{train}}. \quad (99)$$

Define targets

$$y_i(s) = Z_r(\xi_r^{(i)}) a_r(\xi_{r+1}^{(i)}, s) \hat{R}_{r+1}(\xi_{r+1}^{(i)}, s). \quad (100)$$

Fit

$$\hat{R}_r = \arg \min_{R \in \mathcal{H}_r} \sum_{i=1}^{B_r^{\text{train}}} \sum_{\rho=1}^{R_s} \left| R(\xi_r^{(i)}, s^{(\rho)}) - y_i(s^{(\rho)}) \right|^2 + \lambda_r \|R\|_{\mathcal{H}_r}^2. \quad (101)$$

Do not confuse this with NF-iSAM

The regression in Eq. (101) does not learn the posterior density. It learns only a conditional expectation surface needed inside the Slices-style factor construction. This is different from NF-iSAM, which learns normalizing-flow models of clique densities and conditionals.

9 Proposal Guidance from Future Contributions

The original SLICES proposal at level r is essentially

$$q_{r,\text{paper}}(\xi_{r+1} \mid \xi_r) \propto g_r(\xi_r, \xi_{r+1}). \quad (102)$$

But the future contribution toward separator s is

$$g_r(\xi_r, \xi_{r+1})a_r(\xi_{r+1}, s)R_{r+1}(\xi_{r+1}, s). \quad (103)$$

For a fixed separator value,

$$q_{r,\text{smooth}}(\xi_{r+1} \mid \xi_r, s) \propto g_r(\xi_r, \xi_{r+1}) |a_r(\xi_{r+1}, s)R_{r+1}(\xi_{r+1}, s)|. \quad (104)$$

For a proposal reusable over separator values, introduce a relevance density $\rho(s)$ and define

$$q_{r,\text{smooth}}(\xi_{r+1} \mid \xi_r) \propto g_r(\xi_r, \xi_{r+1}) \sqrt{\int \rho(s) [a_r(\xi_{r+1}, s)R_{r+1}(\xi_{r+1}, s)]^2 ds}. \quad (105)$$

To preserve structural support,

$$q_{r,\text{def}}(\xi_{r+1} \mid \xi_r) = (1 - \varepsilon)q_{r,\text{smooth}}(\xi_{r+1} \mid \xi_r) + \varepsilon q_{r,\text{paper}}(\xi_{r+1} \mid \xi_r), \quad (106)$$

$$q_{r,\text{def}}(\xi_{r+1} \mid \xi_r) \geq \varepsilon q_{r,\text{paper}}(\xi_{r+1} \mid \xi_r). \quad (107)$$

10 Computational Reduction: What Can Be Claimed?

10.1 Nested work

If a path has depth H and the paper-style method uses N samples at each level, the worst-case nested sample cardinality behaves as

$$\text{nested sample cardinality} \sim N^H. \quad (108)$$

If the resulting factor is evaluated over R_s separator support points,

$$\text{paper nested cost} \sim N^H R_s. \quad (109)$$

10.2 Smoothed-surface work

Using sparse surface recursion,

$$\text{smooth cost} \sim \sum_{r=0}^{H-1} B_r K_r R_s. \quad (110)$$

Thus the computational saving ratio is approximately

$$\Gamma_{\text{save}} \triangleq \frac{N^H R_s}{\sum_{r=0}^{H-1} B_r K_r R_s} = \frac{N^H}{\sum_{r=0}^{H-1} B_r K_r}. \quad (111)$$

For example, if $N = 200$, $H = 5$, $B_r = 200$, $K_r = 15$, and $R_s = 100$, then

$$N^H = 200^5 = 3.2 \times 10^{11}, \quad (112)$$

$$\sum_{r=0}^4 B_r K_r R_s = 5 \cdot 200 \cdot 15 \cdot 100 = 1.5 \times 10^6. \quad (113)$$

The numerical ratio is enormous, but this is only meaningful if the nested implementation would actually expand the full tree and if $K_r = 15$ is sufficient to approximate the surfaces.

10.3 Honest claim

Open research question

The open research question is not simply whether SMOOTHED SLICES is faster. The precise question is whether the conditional smoothing surfaces R_r have lower effective complexity than the nested sample tree. This should be tested through empirical and theoretical evidence: low rank, sparsity, smoothness, active successor concentration, cache reuse, and surface approximation error.

The method is attractive only when

$$\sum_{r=0}^{H-1} B_r K_r R_s \ll N^H R_s. \quad (114)$$

If $K_r \approx B_{r+1}$ and no compression is possible, then the benefit may shrink. In that case, SMOOTHED SLICES may still reduce variance, but it may not reduce complexity substantially.

11 Relation to NF-iSAM

NF-iSAM uses the Bayes tree to factorize the posterior into clique conditionals:

$$p(\Theta \mid z) = \prod_{C \in \mathcal{C}} p(F_C \mid S_C, z_C), \quad (115)$$

where F_C are frontal variables, S_C are separator variables, and z_C are measurements in and below the clique.

For a clique C , the clique density is built from local factors and child separator densities:

$$p(F_C, S_C \mid z_C) \propto \prod_{q \in \mathcal{C}_C} p(S_q \mid z_q) \prod_{\Theta_{f_i} \subseteq C \text{ or } \Theta_{f_i} \subseteq S_q} f_i(\Theta_{f_i}). \quad (116)$$

A normalizing flow is trained for the clique density. In triangular form,

$$T_C(S_C, F_C) = \left[\begin{array}{c} T_{S_C}(S_C) \\ T_{F_C}(S_C, F_C) \end{array} \right]. \quad (117)$$

Then T_{S_C} models the separator density and T_{F_C} gives a conditional sampler for $p(F_C \mid S_C, z_C)$.

11.1 Key difference

NF-iSAM : learns clique densities/conditionals as normalizing-flow posterior components, (118)

SMOOTHED SLICES : learns or stores conditional expectation surfaces inside a Slices-style factor update. (119)

Thus, SMOOTHED SLICES should not be described as “NF-iSAM with different flows.” It is closer to SLICES but adds a conditional-expectation smoothing layer.

12 Incremental Inference, Caching, and Early Stopping

The SLICES paper proposes incremental re-elimination and a backward-pass early stopping heuristic. The update logic is:

$$\text{new measurement} \rightarrow \text{identify affected variables/subtree} \rightarrow \text{re-eliminate affected section} \rightarrow \text{perform backward pass} \quad (120)$$

The backward early stopping metric is MMD. For a variable θ_j , let $\hat{P}^{(k)}(\theta_j \mid D)$ be the new marginal and $\hat{P}^{(k-1)}(\theta_j \mid D)$ be the cached old marginal. Stop the backward pass when

$$\text{MMD}\left(\hat{P}^{(k)}(\theta_j \mid D), \hat{P}^{(k-1)}(\theta_j \mid D)\right) < \vartheta. \quad (121)$$

The hyperparameters are

$$N_M = \text{number of samples used for MMD}, \quad \vartheta = \text{stopping threshold}. \quad (122)$$

For SMOOTHED SLICES, the cache becomes

$$\mathcal{C}_k = \left\{ \hat{f}_{\text{new}}^{(j,k)}, \hat{P}^{(k)}(\theta_j \mid S_j, D_j), \hat{R}_r^{(k)} \right\}_{j,r \in \text{affected region}}. \quad (123)$$

The surface update is

$$\hat{R}_r^{(k)} = \begin{cases} \text{Update}\left(\hat{R}_r^{(k-1)}, \text{new factors}\right), & r \in \mathcal{T}_{\text{aff}}, \\ \hat{R}_r^{(k-1)}, & r \notin \mathcal{T}_{\text{aff}}. \end{cases} \quad (124)$$

13 Plaza1 and Plaza2 Case-Study Integration

The case-study purpose is to compare three methods:

$$\mathcal{M} = \{\text{NF-iSAM}, \text{SLICES}, \text{SMOOTHED SLICES}\}. \quad (125)$$

The datasets are Plaza1 and Plaza2, both involving range and odometry data with unknown landmark locations. The demonstration should emphasize recovery of full posterior structure rather than only MAP accuracy.

13.1 Required figures

The app and report should generate the following figure families:

1. **Trajectory and landmark posterior samples:** plot ground truth, odometry, posterior samples, empirical mean, and landmark beliefs.
2. **Marginal KDE plots:** for selected landmarks, plot x and y marginal distributions across time, similar to the paper-style Plaza figures.
3. **RMSE over incremental steps:**

$$\text{RMSE}_k = \sqrt{\frac{1}{d} \|\hat{\mu}_k - \theta_k^{\text{GT}}\|^2}. \quad (126)$$

4. **Runtime per increment and accumulated runtime.**

5. **MMD over time:** compare distributional shape, not only point estimate.
6. **Variable-elimination slider panels:** show which variable is eliminated, its separator, removed factors, and new factor.
7. **NF-iSAM Bayes-tree process panel:** show changed subtree, leaf-to-root training, cached flows, and root-to-leaf sampling.
8. **Smoothed Slices surface panel:** show R_r surfaces, active successor sets, compression rank, surface error proxies, and cache reuse.
9. **Complexity research plots:**

$$\text{nested count} \sim N^H, \quad \text{surface count} \sim \sum_r B_r K_r R_s. \quad (127)$$

13.2 Research diagnostics

For each dataset, increment, method, and random seed, log

$$\mathcal{D}_{\log} = \{k, \text{dataset}, \text{method}, N, H, B_r, K_r, R_s, \text{factor evals}, \text{runtime}, \text{MMD}, \text{RMSE}, \text{cache hit ratio}\}. \quad (128)$$

For SMOOTHED SLICES, additionally log

$$\mathcal{D}_{\text{surface}} = \{\text{rank}(\mathbf{R}_r), \text{sparsity}(\mathbf{R}_r), \text{smoothness}(\mathbf{R}_r), \text{reuse}(\mathbf{R}_r), \text{residual}(\mathbf{R}_r)\}. \quad (129)$$

These diagnostics are what allow the app to study the open question of whether R_r is truly lower-complexity than the nested sample tree.

14 MATLAB App Implementation Requirements

The MATLAB app should be designed as a comparative research demonstrator, not merely as a plotting GUI. It should run the same case study through all three methods and expose the internal process.

14.1 LaTeX interpreter requirement

Every UI label, axis label, title, legend, table header where MATLAB supports it, annotation, exported figure title, and diagnostic report should use LaTeX formatting where possible. For example:

```
title(app.UIAxesPosterior, '$\widehat{P}(x,y\mid D_k)$', 'Interpreter','latex');
xlabel(app.UIAxesPosterior, '$x_{[m]}$', 'Interpreter','latex');
ylabel(app.UIAxesPosterior, '$y_{[m]}$', 'Interpreter','latex');
legend(app.UIAxesPosterior, {'$\mathrm{GT}$', '$\mathrm{samples}$', '$\widehat{\mu}$'},
    ...
    'Interpreter','latex');
```

14.2 Working-directory hierarchy

The project should be structured as follows:

```

SmoothedSlicesProject/
  App/
    SmoothedSlicesApp.mlapp
    startup_app.m
  Backend/
    Methods/
      run_slices.m
      run_nf_isam.m
      run_smoothed_slices.m
    Factors/
      range_factor.m
      odometry_factor.m
      ambiguous_range_factor.m
    Graph/
      build_factor_graph.m
      variable_elimination_order.m
      build_bayes_tree.m
    SmoothingSurfaces/
      build_supports.m
      update_surface_direct.m
      update_surface_basis.m
      update_surface_regression.m
      active_successors.m
    Metrics/
      compute_rmse.m
      compute_mmd.m
      count_factor_evals.m
      surface_diagnostics.m
    Plotting/
      plot_posterior_samples.m
      plot_marginal_kde.m
      plot_elimination_process.m
      plot_bayes_tree_process.m
      plot_surface_panels.m
      export_all_figures.m
  Data/
    Plaza1/
    Plaza2/
  Results/
    Runs/
    Figures/
    Diagnostics/
    Reports/
  Instructions/
    slices_exact_implementation.docx
    nf_isam_exact_implementation.docx
    smoothed_slices_exact_implementation.docx
    plaza_case_study_showcase_manual.docx
    smoothed_slices_research_tool_instructions.docx
  Literature/
    A_Slices_Perspective_IROS_2024.pdf
    NF_iSAM_TRO_2023.pdf
    iSAM2_reference.pdf
    GTSAM_or_Plaza_dataset_notes.pdf

```

14.3 Figure generator

Add a button called **Export All Figures and Diagnostics**. It should:

1. save every app axes as PNG, PDF, and FIG;
2. save diagnostics as MAT and CSV;
3. create a timestamped folder;
4. export method configuration as JSON or MAT;
5. generate a summary PDF/HTML report if possible.

14.4 Process sliders

Add sliders for process visualization:

- **Increment slider:** selects time step k .
- **Elimination slider for Slices and Smoothed Slices:** selects elimination index j and displays θ_j , S_j , removed factors, sampling factor, and new factor.
- **Bayes-tree slider for NF-iSAM:** selects clique C , showing F_C , S_C , child separator densities, learned flow status, and top-down sampling state.
- **Surface-depth slider for Smoothed Slices:** selects r and displays $\mathbf{R}_r$, active successors, rank/sparsity, and approximation residual.

15 Validation and Acceptance Criteria

15.1 Mathematical acceptance

A correct implementation must satisfy:

$$\hat{f}_{\text{new,slices}}(S_j \mid D_j) \approx \eta_j^{-1} \int \prod_{f_i \in F_{j-1}(\theta_j)} f_i(\Theta_i) d\theta_j, \quad (130)$$

$$\hat{R}_r(\xi_r, s) \approx \int g_r(\xi_r, \xi_{r+1}) a_r(\xi_{r+1}, s) \hat{R}_{r+1}(\xi_{r+1}, s) d\xi_{r+1}, \quad (131)$$

$$\hat{f}_{\text{new,smooth}}(s) \approx \frac{c_0}{N_0} \sum_n b_0(\xi_0^{(n)}, s) \hat{R}_0(\xi_0^{(n)}, s). \quad (132)$$

15.2 Research acceptance

The tool should not merely output that SMOOTHED SLICES is faster. It must generate evidence:

$$\Gamma_{\text{save}}(k) = \frac{\text{measured factor evaluations of paper-like nested baseline at } k}{\text{measured factor evaluations of Smoothed Slices at } k}, \quad (133)$$

$$\Delta_{\text{MMD}}(k) = \text{MMD}_{\text{SMOOTHED SLICES}}(k) - \text{MMD}_{\text{SLICES}}(k), \quad (134)$$

$$\Delta_{\text{RMSE}}(k) = \text{RMSE}_{\text{SMOOTHED SLICES}}(k) - \text{RMSE}_{\text{SLICES}}(k). \quad (135)$$

A convincing result is not only low runtime. It is low runtime with comparable or improved posterior recovery.

16 Failure Modes and Honest Limitations

Important limitations

1. The smoothing property reorganizes the computation; it does not remove the mathematical integral.
2. If R_r is high-complexity, dense, non-smooth, or non-compressible, there may be little computational gain.
3. Approximate surfaces can introduce bias.
4. Low RMSE is not enough for non-Gaussian posterior recovery; MMD, KDE shape, mode recovery, and posterior samples must also be inspected.
5. A flow or neural model used as a proposal must not be mistaken for the posterior itself unless the algorithm is explicitly NF-ISAM.
6. The research tool must record factor evaluation counts, not only wall-clock time, because wall-clock time depends on MATLAB implementation details.

For approximate surfaces,

$$\hat{R}_r(\xi_r, s) = R_r(\xi_r, s) + \delta_r(\xi_r, s). \quad (136)$$

In the simple R_1 case,

$$\hat{f}_{\text{smooth}}^{(2)}(x_2) - \tilde{f}_{\text{new}}^{(2)}(x_2) = \frac{c_1}{N_1} \sum_{n=1}^{N_1} f(x_1^{(n)}, x_2) \delta_1(x_1^{(n)}, x_2). \quad (137)$$

By Cauchy-Schwarz,

$$\left| \hat{f}_{\text{smooth}}^{(2)}(x_2) - \tilde{f}_{\text{new}}^{(2)}(x_2) \right| \leq |c_1| \sqrt{\frac{1}{N_1} \sum_{n=1}^{N_1} f^2(x_1^{(n)}, x_2)} \sqrt{\frac{1}{N_1} \sum_{n=1}^{N_1} \delta_1^2(x_1^{(n)}, x_2)}. \quad (138)$$

Thus the method is useful only when

$$\text{surface approximation error} < \text{inner nested Monte Carlo error saved}. \quad (139)$$

17 The Clean Final Story

The final story should be stated like this:

Final story

SLICES proves that graph structure can make difficult variables sampleable during nonparametric variable elimination. The nested sampling step that appears around the $x_1 \rightarrow l_1 \rightarrow x_2$ example can be rewritten as a conditional expectation. SMOOTHED SLICES keeps the same Slices structure, but replaces repeated nested sample branches by conditional smoothing surfaces $R_r(\xi_r, s)$ that summarize expected future factor contributions. These surfaces are cached, updated, visualized, and tested for lower effective complexity in Plaza1/Plaza2 and synthetic case studies.

In a single line:

$$\text{sampleability path} \longrightarrow \text{tower of conditional expectations} \longrightarrow \text{cached smoothing surfaces} \quad (140)$$

A Reconstructed Prompt and Deliverable Log

The following list records the main user requests that shaped the method and artifacts:

1. Request to write a detailed theoretical direction for a system/dynamic-model idea in combustion, which established the user’s preference for state-of-the-art documents with equations, pipeline diagrams, and citations.
2. Request to compare and understand NF-iSAM and SLICES deeply, especially how NF-iSAM learns clique conditionals and how SLICES avoids learned intermediate reconstructions.
3. Request to identify whether the SLICES paper’s Eq. (16)-type construction contains a place for innovation.
4. Development of overlap-aware mixture slices and multiple-importance bridge slices, where future factors are used to guide proposals rather than merely evaluated after sampling.
5. Development of the more important idea: reinterpret the nested sampling step through the smoothing property of conditional expectation.
6. Request to avoid the word “message” and instead use “conditional smoothing surface” or “recursive smoothing surface.”
7. Request to emphasize implementation cardinalities in blue and write them as $|\mathcal{X}_r|$, $|\mathcal{S}|$, and $|\mathcal{N}_r|$.
8. Creation of the RECURSIVE CONDITIONAL-SMOOTHING SLICES formulation note in Overleaf/PDF form.
9. Request to design a MATLAB app demonstrating the approach, initially called SMOOTHED SLICES.
10. Request to generate two implementation files for the exact author-specified methods: SLICES and NF-iSAM.
11. Request to generate a third implementation file for SMOOTHED SLICES.
12. Request to strengthen the MATLAB app specification: all LaTeX interpreters, all-figures export button, diagnostics, working-directory hierarchy with **Instructions/** and **Literature/**, process sliders, and comparative purpose for all three methods.
13. Request to create a Plaza1/Plaza2 showcase manual for comparing the methods on classical SLAM problems.
14. Request to explain the intuition of SMOOTHED SLICES concisely in chat.
15. Request to emphasize how conditional-expectation surfaces insert future factor information and how implementation differs from SLICES.
16. Question: in words, how many calculations can it reduce? The answer: potentially from nested tree-like work N^H to surface recursion $\sum_r B_r K_r R_s$, but only if the surfaces are lower-complexity.
17. Request to create an AI instruction sheet to research this open computational-reduction question.
18. Request to export the most detailed report on the entire discussion.
19. Current request: create a longer Overleaf report focusing on formulations, the intuition, and the convergence toward the smoothing property of conditional expectation.

B Glossary

Symbol / term	Meaning
$G = (F, \Theta, E)$	Factor graph with factors, variables, and edges.
O	Variable elimination order.
θ_j	Variable eliminated at step j .
S_j	Separator/Markov blanket variables remaining after eliminating θ_j .
D_j	Local data represented by removed factors up to step j ; in approximate SLICES, may include previous $\hat{f}_{\text{new}}$.
f_{new}	New separator factor produced by marginalizing an eliminated variable.
q_j	Proposal induced by the selected sampling factor.
$R_1(x_1, x_2)$	First conditional smoothing surface; expected contribution of l_1 to separator x_2 given x_1 .
$R_r(\xi_r, s)$	Recursive smoothing surface; expected downstream factor contribution toward separator s conditioned on path variable ξ_r .
$\mathcal{X}_r$	Finite online support for ξ_r .
$\mathcal{S}$	Finite online support for separator s .
$\mathcal{N}_r(a)$	Active successor set for support point $\xi_r^{(a)}$.
B_r	Number of support points in $\mathcal{X}_r$.
R_s	Number of separator support points.
K_r	Number of active successors per support point.
NF-iSAM	Normalizing-flows for incremental smoothing and mapping; learns Bayes-tree clique conditionals.
SLICES	Direct slice-based nonparametric inference through variable elimination.
SMOOTHED SLICES	Proposed extension: Slices plus recursive conditional-expectation surfaces.

C References

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